

PVsyst - Simulation report

Standalone system

Project: UAS RIDWAN DKE

Variant: New simulation variant

Standalone system with batteries

System power: 2200 Wp

Malang - Indonesia

**PVsyst V7.4.7**

VC1, Simulation date:
06/07/26 00:20
with V7.4.7

Project summary**Geographical Site**

Malang
Indonesia

Situation

Latitude -7.98 °S
Longitude 112.63 °E
Altitude 440 m
Time zone UTC+7

Project settings

Albedo 0.20

Weather data

Malang
Meteonorm 8.1 (2016-2021), Sat=100% - Synthetic

System summary**Standalone system****PV Field Orientation**

Fixed plane
Tilt/Azimuth 30 / 0 °

Standalone system with batteries**User's needs**

Daily household consumers
Monthly Specifications
Average 0.9 kWh/Day

System information**PV Array**

Nb. of modules 4 units
Pnom total 2200 Wp

Battery pack

Technology Lithium-ion, LFP
Nb. of units 7 units
Voltage 48 V
Capacity 700 Ah

Results summary

Useful energy from solar	225.36 kWh/year	Specific production	102 kWh/kWp/year	Perf. Ratio PR	5.14 %
Missing Energy	85.38 kWh/year	Available solar energy	3754.06 kWh/year	Solar Fraction SF	72.54 %
Excess (unused)	3482.32 kWh/year				

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General parameters

Standalone system

PV Field Orientation

Orientation

Fixed plane

Tilt/Azimuth 30 / 0 °

User's needs

Daily household consumers

Monthly Specifications

Average 0.9 kWh/Day

Standalone system with batteries

Sheds configuration

No 3D scene defined

Models used

Transposition Perez

Diffuse Perez, Meteor norm

Circumsolar separate

PV Array Characteristics

PV module

Manufacturer Trina Solar

Model TSM-DE19-550Wp Vertex

(Original PVsyst database)

Unit Nom. Power 550 Wp

Number of PV modules 4 units

Nominal (STC) 2200 Wp

Modules 2 string x 2 In series

At operating cond. (50°C)

Pmpp 2013 Wp

U mpp 57 V

I mpp 35 A

Controller

Universal controller

Technology MPPT converter

Temp coeff. -5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies 97.0 / 95.0 %

Total PV power

Nominal (STC) 2.20 kWp

Total 4 modules

Module area 10.5 m²

Battery

Manufacturer ZTE

Model ZXESM R311, 100Ah

Technology Lithium-ion, LFP

Nb. of units 7 in parallel

Discharging min. SOC 10.0 %

Stored energy 31.4 kWh

Battery Pack Characteristics

Voltage 48 V

Nominal Capacity 700 Ah (C10)

Temperature Fixed 20 °C

Battery Management control

Threshold commands as SOC calculation

Charging SOC = 0.96 / 0.80

Discharging SOC = 0.10 / 0.35

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const) 20.0 W/m²KUv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 27 mΩ

Loss Fraction 1.5 % at STC

Serie Diode Loss

Voltage drop 0.7 V

Loss Fraction 1.1 % at STC

Module Quality Loss

Loss Fraction -0.4 %

Module mismatch losses

Loss Fraction 0.5 % at MPP

Strings Mismatch loss

Loss Fraction 0.1 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



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Detailed User's needs

Daily household consumers, Monthly Specifications, average = 0.9 kWh/day

January and February

	Nb.	Power	Use	Energy	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day		W	Hour/day	Wh/day
BEBAN	1	10000/lamp	1.0	10000				
Stand-by consumers			24.0	24				
Total daily energy				10024				0

March and April

	Nb.	Power	Use	Energy	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day		W	Hour/day	Wh/day
Total daily energy				0				0

May and June

	Nb.	Power	Use	Energy	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day		W	Hour/day	Wh/day
Total daily energy				0				0

July and August

	Nb.	Power	Use	Energy	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day		W	Hour/day	Wh/day
Total daily energy				0				0

September and October

	Nb.	Power	Use	Energy	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day		W	Hour/day	Wh/day
Total daily energy				0				0

November and December

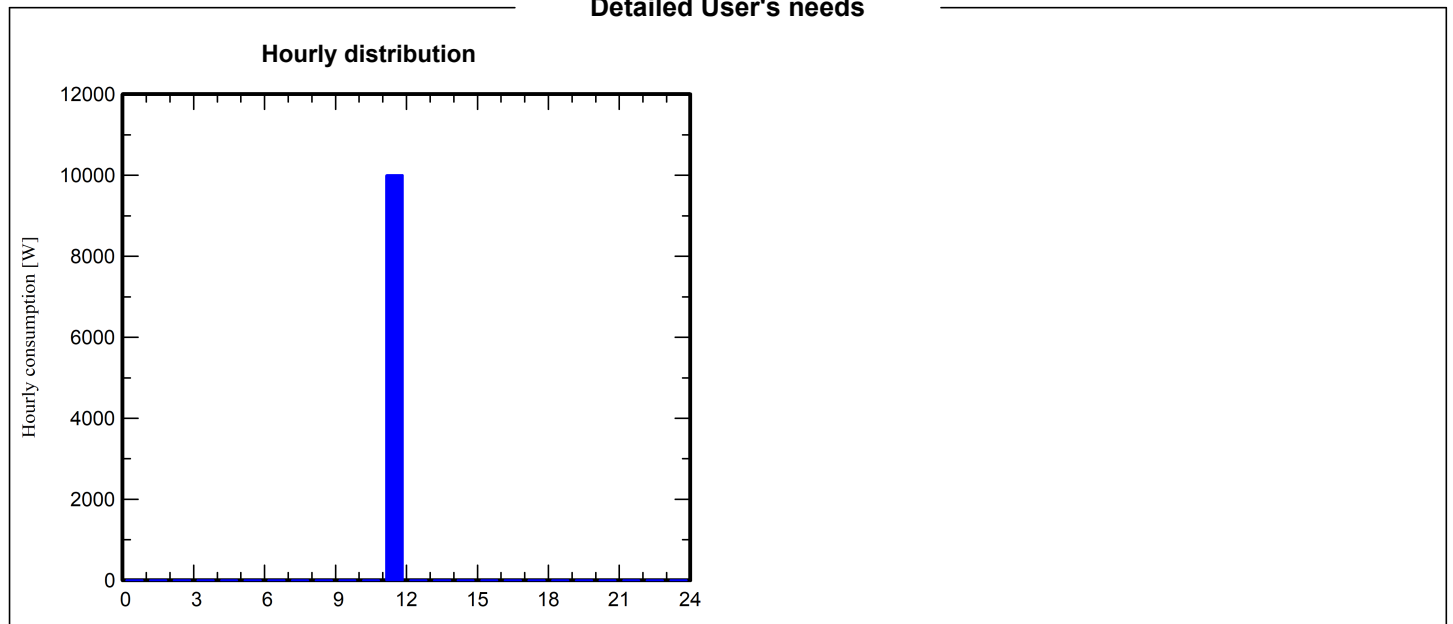
	Nb.	Power	Use	Energy	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day		W	Hour/day	Wh/day
Total daily energy				0				0



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Detailed User's needs





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Main results

System Production

Useful energy from solar	225.36 kWh/year
Available solar energy	3754.06 kWh/year
Excess (unused)	3482.32 kWh/year

Perf. Ratio PR	5.14 %
Solar Fraction SF	72.54 %

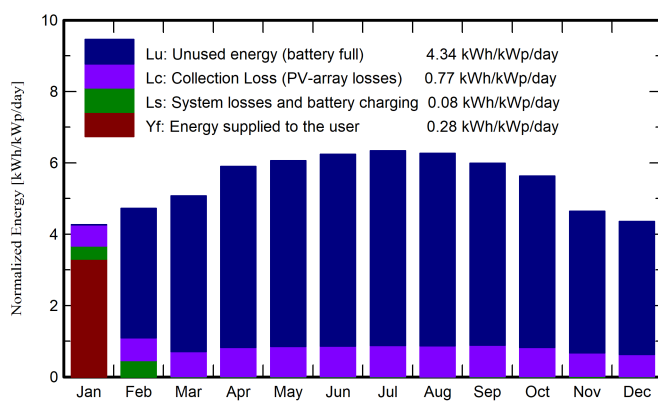
Loss of Load

Time Fraction	0.0 %
Missing Energy	85.38 kWh/year

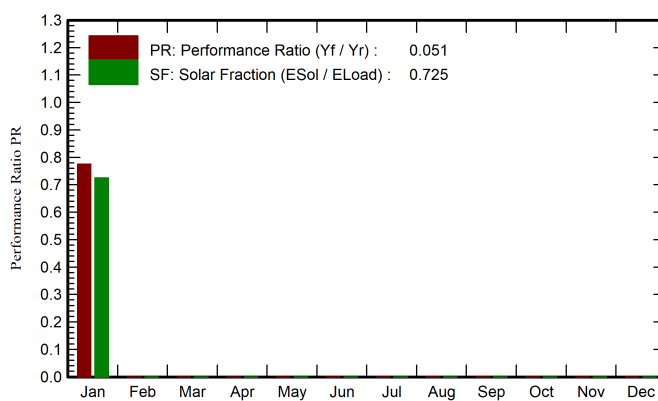
Battery aging (State of Wear)

Cycles SOW	99.8 %
Static SOW	93.3 %
Battery lifetime	15.0 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	GlobEff kWh/m ²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	167.6	127.1	239.4	0.0	85.38	225.4	310.7	0.725
February	154.2	128.1	249.7	222.8	0.00	0.0	0.0	0.000
March	165.6	153.7	298.1	297.6	0.00	0.0	0.0	0.000
April	167.6	173.5	334.7	334.2	0.00	0.0	0.0	0.000
May	161.6	185.0	355.3	354.7	0.00	0.0	0.0	0.000
June	152.6	184.7	355.7	355.2	0.00	0.0	0.0	0.000
July	164.6	193.7	372.8	372.2	0.00	0.0	0.0	0.000
August	175.8	191.1	368.0	367.4	0.00	0.0	0.0	0.000
September	181.0	175.9	336.8	336.2	0.00	0.0	0.0	0.000
October	195.9	169.8	327.2	326.8	0.00	0.0	0.0	0.000
November	172.4	134.1	262.0	261.3	0.00	0.0	0.0	0.000
December	175.8	129.4	254.5	253.9	0.00	0.0	0.0	0.000
Year	2034.7	1946.1	3754.1	3482.3	85.38	225.4	310.7	0.725

Legends

GlobHor	Global horizontal irradiation
GlobEff	Effective Global, corr. for IAM and shadings
E_Avail	Available Solar Energy
EUnused	Unused energy (battery full)
E_Miss	Missing energy

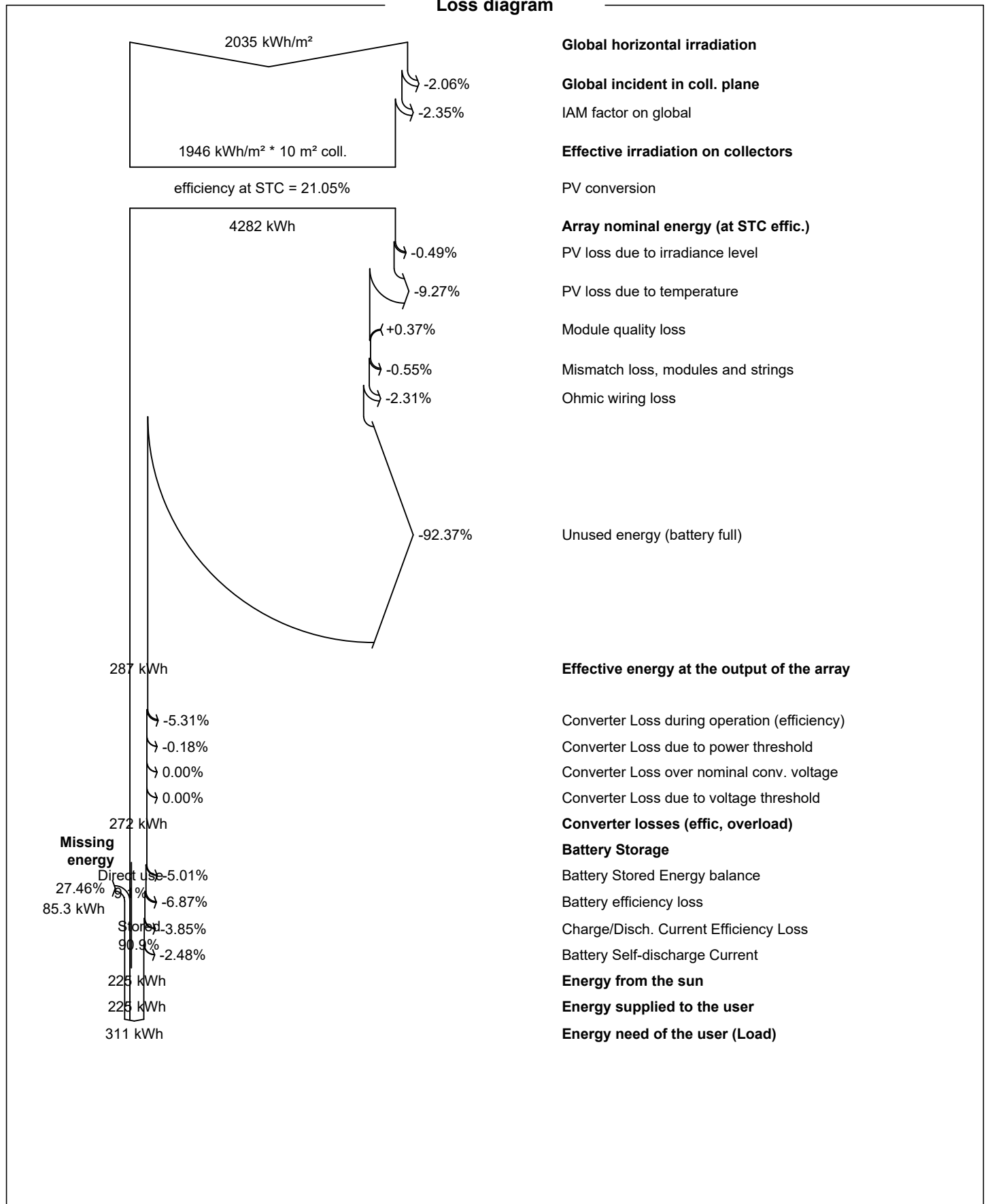
E_User	Energy supplied to the user
E_Load	Energy need of the user (Load)
SolFrac	Solar fraction (EUsed / ELoad)



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Loss diagram





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Predef. graphs

Daily Input/Output diagram

